SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1712**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I R. Sandhya Rani(192372378) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1. Smart Home Automation System – Intelligent Agents

1. Simple Reflex Agent

- Acts only on the current input using predefined rules.
- **Example:**
  - If motion is detected → Turn ON lights.
  - If no motion → Turn OFF lights.
- **Advantage:** Fast and simple.
- **Limitation:** Does not remember past events or learn user preferences.

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2. Model-Based Agent

- Maintains an internal model of the home environment.
- Uses previous and current information to make decisions.
- **Example:**
  - Remembers that the bedroom is occupied even if motion temporarily stops.
  - Adjusts temperature based on current room conditions.
- **Advantage:** Handles partially observable environments.
- **Limitation:** Does not optimize for user preferences.

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3. Goal-Based Agent

- Selects actions to achieve specific goals.
- **Example Goals:**
  - Maintain room temperature at **24°C**.
  - Keep the home secure.
  - Reduce unnecessary energy usage.
- **Advantage:** Chooses actions that help achieve desired objectives.
- **Limitation:** Does not compare multiple solutions based on overall benefit.

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4. Utility-Based Agent

- Chooses the action that provides the highest overall benefit (utility).
- Considers comfort, energy efficiency, and security together.
- **Example:**
  - Keeps lights dim instead of fully ON to save energy.
  - Balances air conditioner usage with electricity consumption.
  - Activates security alerts only when necessary.
- **Advantage:** Produces the best overall performance.
- **Limitation:** More complex to design.

Comparison

Agent Type	Function in Smart Home	Advantage	Limitation
Simple Reflex	Turns devices ON/OFF based on current input	Fast and easy	No memory or learning
Model-Based	Uses current and previous information	Handles changing conditions	No optimization
Goal-Based	Achieves predefined goals	Goal-oriented decisions	Ignores overall utility
Utility-Based	Maximizes comfort, security, and energy efficiency	Best overall decision	More complex

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Best Approach (Hybrid Agent)

A **hybrid approach combining Model-Based and Utility-Based agents** is the most effective because it:

- Learns user preferences over time.
- Uses previous and current sensor data.
- Optimizes comfort, energy efficiency, and security.
- Makes intelligent decisions in real time.

Conclusion

A **Model-Based + Utility-Based Hybrid Agent** is the best choice for a smart home automation system because it remembers environmental conditions, adapts to user behavior, and selects actions that maximize comfort, minimize energy consumption, and maintain home security.

2. **2. Robot Navigation in an Unknown Maze**

(a) Uniform Cost Search (UCS)

- UCS expands the node with the **lowest path cost** first.
- It guarantees the **least-cost path** if all costs are positive.
- Suitable when different paths have different movement costs.

Advantages:

- Finds the optimal path.
- Works well with varying path costs.

Disadvantages:

- High time complexity.
- Requires large memory to store explored nodes.

(b) Iterative Deepening Search (IDS)

- IDS performs repeated Depth-First Search with increasing depth limits.
- It finds the shallowest solution while using less memory.

Advantages:

- Low memory usage.
- Complete and suitable for unknown-depth problems.

Disadvantages:

- Repeats node expansions.
- Slower than BFS when the solution is deep.

(c) Time and Space Complexity

Algorithm	Time Complexity	Space Complexity	Optimal
Uniform Cost Search (UCS)	($O(b^{1+\lfloor C^*/\epsilon \rfloor})$)	Same as time (high)	Yes
Iterative Deepening Search (IDS)	($O(b^d)$)	($O(b \times d)$)	Yes (for equal step costs)

Where:

- **b** = branching factor
- **d** = depth of solution
- **C*** = optimal path cost
- **ε** = minimum step cost

(d) Intelligent Agents

Simple Reflex Agent

- Acts only on current perception.

- Cannot remember previous paths.
- Not suitable for maze navigation.

Model-Based Agent

- Maintains an internal map of explored paths.
- Avoids revisiting blocked or explored areas.
- Suitable for unknown environments.

Goal-Based Agent

- Plans actions to reach the exit.
- Selects paths that move toward the goal.

Utility-Based Agent

- Chooses paths that maximize overall benefit (minimum cost, minimum risk, shortest time).

(e) Best Combination

Recommended: Model-Based Goal-Based Agent + Uniform Cost Search (UCS)

Justification:

- The **Model-Based Agent** remembers explored locations and updates its map.
- The **Goal-Based Agent** focuses on reaching the exit.
- **Uniform Cost Search** guarantees the minimum-cost path even when movement costs vary.

Conclusion

A **Model-Based Goal-Based Agent using Uniform Cost Search (UCS)** is the most effective solution because it can learn the maze layout, make informed decisions, avoid revisiting explored paths, and find the optimal path to the exit. If all moves have equal cost and memory is limited, **Iterative Deepening Search (IDS)** is a good alternative.